



Laboratory Assessment V

PEC-2 Laboratory: Full Stack Development - BIT26PE05

To test and validate the functionality of a RESTful API

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AIM

To test and validate the functionality of a RESTful API for managing employee records using Postman by performing CRUD operations and query-based retrieval.

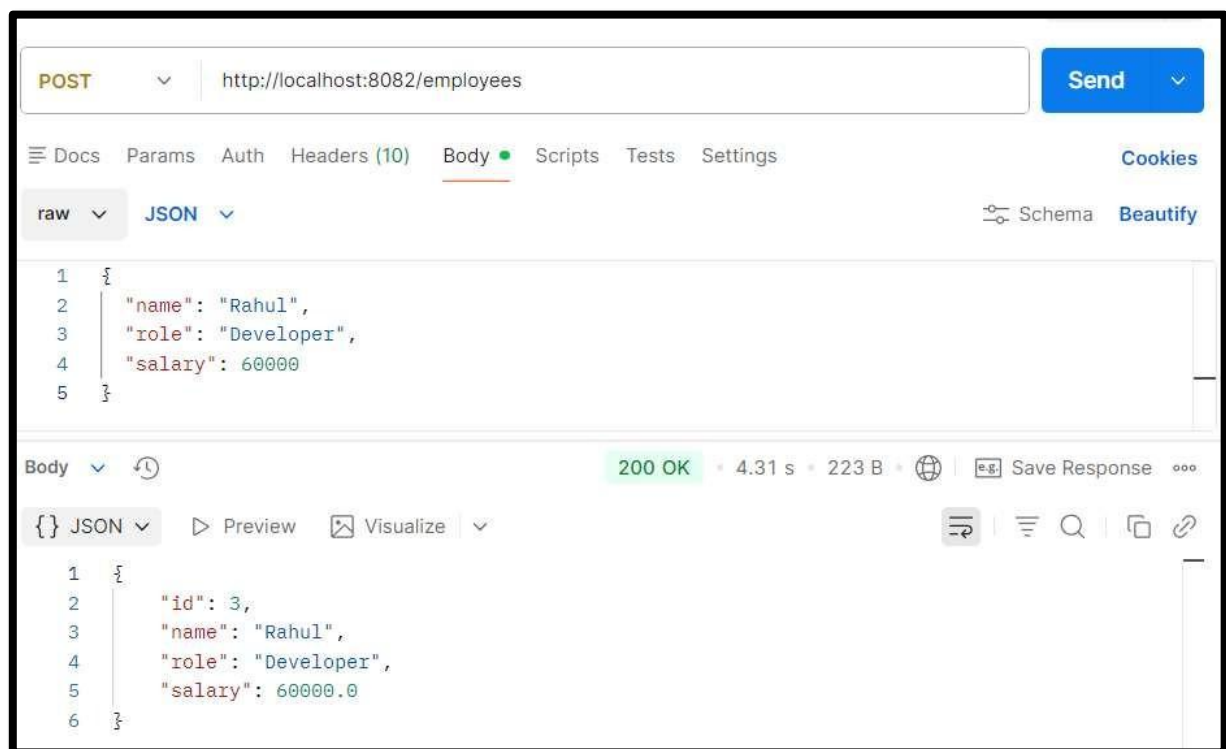
OBJECTIVES

- To verify all CRUD operations of the Employee REST API
- To test API endpoints using Postman
- To validate request and response handling (JSON format)
- To ensure correct HTTP status codes are returned
- To perform query-based search (e.g., by employee name or department)
- To identify potential errors and ensure API reliability

Add New Employee

Method: POST

Endpoint: /employees



```
select *from employees;
```

```
select *from employees;
```

ID	NAME	ROLE	SALARY
1	John	Developer	100000.0
3	Rahul	Developer	60000.0

(2 rows, 9 ms)

Retrieve All Employees

Method: GET

Endpoint: /employees

GET <http://localhost:8082/employees>

Docs Params Auth Headers (10) Body Scripts Tests Settings

raw JSON

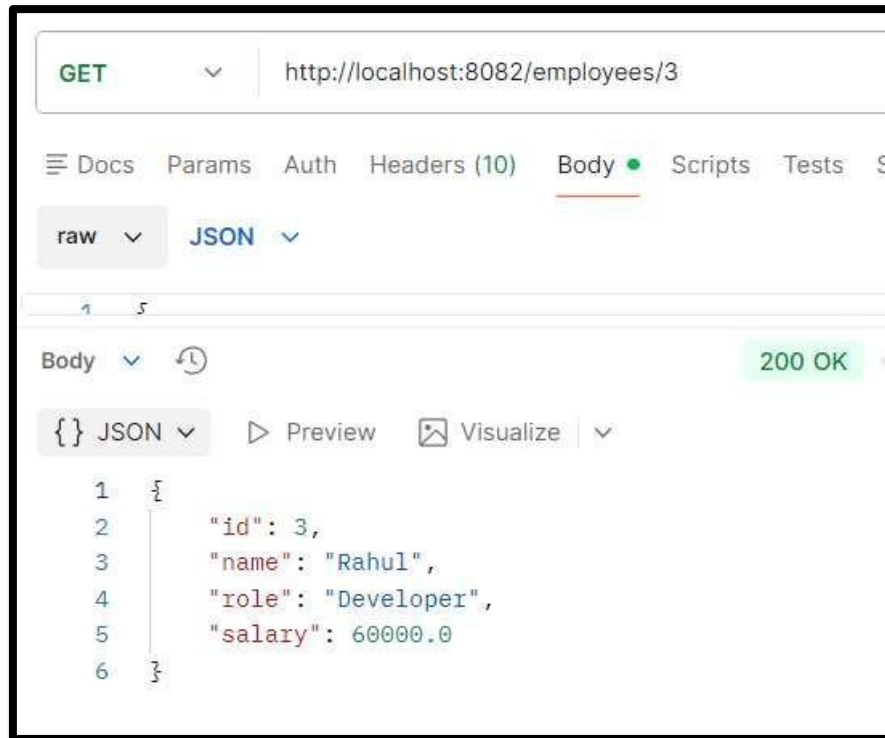
Body 200 OK • 1.71 s

```
{  
  "id": 1,  
  "name": "John",  
  "role": "Developer",  
  "salary": 100000.0  
},  
{  
  "id": 3,  
  "name": "Rahul",  
  "role": "Developer",  
  "salary": 60000.0  
}
```

Retrieve Employee by ID

Method: GET

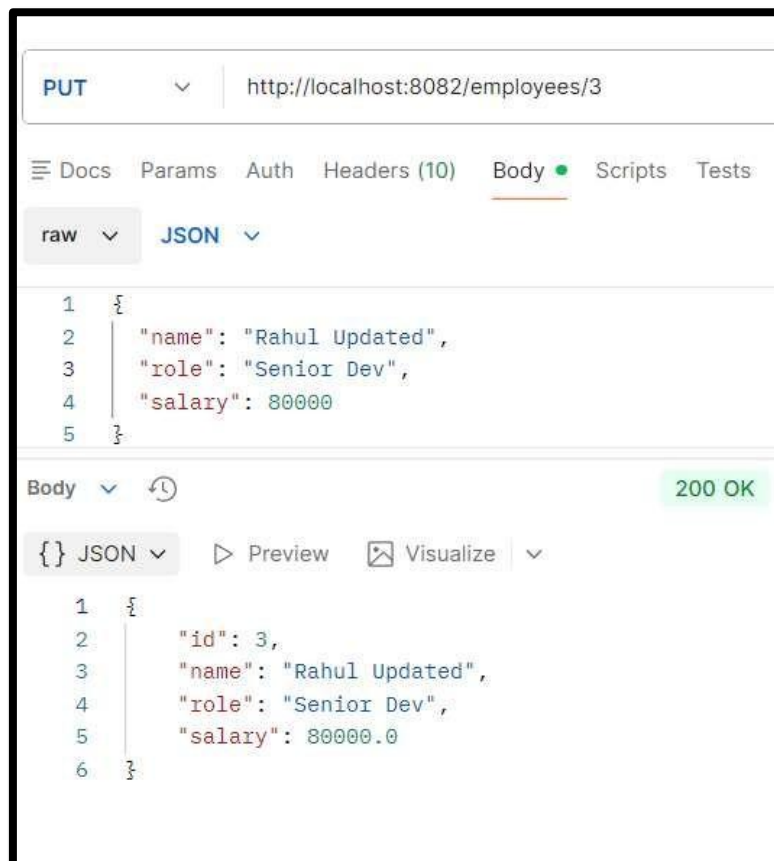
Endpoint: /employees/{id}



Update Employee Details

Method: PUT

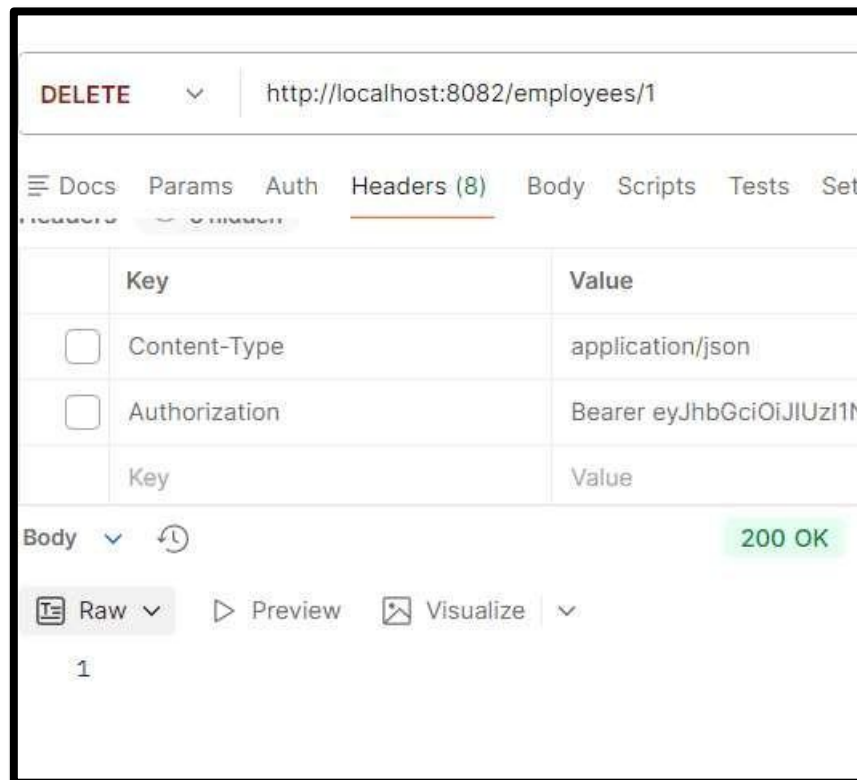
Endpoint: /employees/{id}



Delete Employee

Method: DELETE

Endpoint: /employees/{id}



DELETE ▼ http://localhost:8082/employees/1

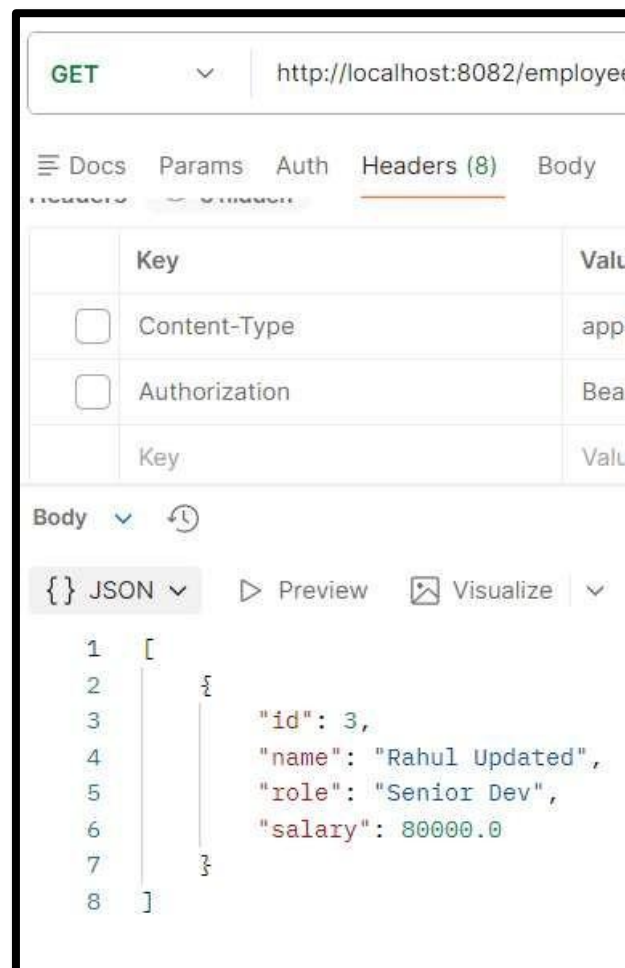
Docs Params Auth Headers (8) Body Scripts Tests Set

	Key	Value
☐	Content-Type	application/json
☐	Authorization	Bearer eyJhbGciOiJIUzI1Ni...
	Key	Value

Body ▼ 🔄 200 OK

☒ Raw ▼ ▶ Preview 🖼️ Visualize ▼

1



GET ▼ http://localhost:8082/employees

Docs Params Auth Headers (8) Body

	Key	Value
☐	Content-Type	app
☐	Authorization	Bea
	Key	Valu

Body ▼ 🔄

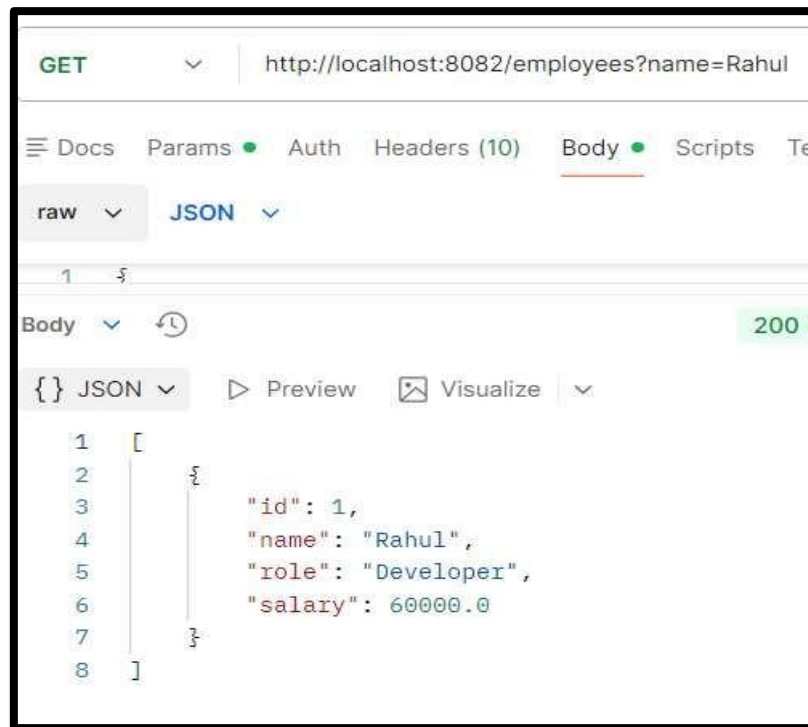
☒ JSON ▼ ▶ Preview 🖼️ Visualize ▼

```
1  [
2    {
3      "id": 3,
4      "name": "Rahul Updated",
5      "role": "Senior Dev",
6      "salary": 80000.0
7    }
8  ]
```

Search Employee by Name (Query Param)

Method: GET

Endpoint: /employees?name=John



CONCLUSION

The Postman testing successfully validated the functionality of the Employee RESTful API by executing CRUD operations and query-based retrieval. The API demonstrated correct handling of requests, appropriate HTTP status codes, and proper data exchange in JSON format. This testing process highlights the importance of systematic API validation to ensure reliability, accuracy, and robustness in real-world applications.